



Change One Script

Quick facilitation notes & answer key

Goal

Students ALTER one block in one of two concurrent scripts and DESCRIBE how that change affects the combined outcome — the heart of Grade-2 C3.2 ('describe how changes to the code affect the outcomes'). Changing Pixel's 'back 1' to 'back 2' moves Pixel from 4 to 3, flipping the result from 'do not meet' (Pixel ahead) to 'they meet on 3'.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two counters on a 0-5 path; run the BEFORE version, then physically change Pixel's move and run the AFTER version so the change is visible.

Run it (about 20 min)

1. I DO: run the BEFORE version — 'Bit 0 forward 3 → 3; Pixel 5 back 1 → 4. They do NOT meet; Pixel is ahead.'
2. WE DO: make the change together — Pixel's 'back 1' becomes 'back 2'. Predict: where will Pixel land now?
3. YOU DO: students run the AFTER version (Pixel → 3), see they now MEET, and write one sentence describing the effect.
4. Connect: 'One small change to one script changed the whole result — that's why coders test after every change.'

Answer key (modelled by the run-gate)

Ex 1 (before) — Bit: 0 → forward 3 → 3. Pixel: 5 → back 1 → 4. They do NOT meet; Pixel (4) is ahead of Bit (3).

Ex 2 (after) — Pixel's new script: 5 → back 2 → 3. Bit still 3. Now both are on 3 → YES, they MEET.

Ex 3 (describe) — sample: 'Changing Pixel to back 2 made Pixel land on 3 instead of 4, so now Pixel meets Bit instead of being ahead.' Accept any answer that links the changed move to the changed outcome.

Interaction note: only Pixel's script changed; Bit's result stayed the same, but the COMBINED outcome (meet vs not) flipped.

Watch for & differentiate

Common error: re-running Bit differently — remind students only Pixel's script changed; Bit is unchanged.

Common error: stating the new landing without describing the EFFECT — push for 'now they meet' / 'Pixel is no longer ahead'.

Simplify: just compare Pixel's two landings (4 then 3) and ask what that does to 'do they meet?'

Extend: ask what change to Bit's script would make them NOT meet again.

Success indicator: the child reports meet=NO before and meet=YES after, and names the changed move as the cause.